Does AI-Assisted Writing Resurrect Retracted Science?
A Population-Level Study of Zombie Citations

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Abstract

Large language models trained before a retraction cannot know about it, and audits of chatbots show them drawing on retracted articles while flagging the retraction less than half the time. Whether that failure reaches the published literature is unknown. We measure it across 204.9 million citing works. Combining the Retraction Watch database with a frozen OpenAlex snapshot, we track citations to papers already retracted when they were cited, which we call zombie citations. Three designs answer the question in three ways, and their disagreement is the result. The population rate rises by a fifth at the release of ChatGPT, exceeding all 36 placebo breaks and concentrating in computational fields. Papers carrying phrases such as "as an AI language model" verbatim, where the filter reaches 70% precision, cite retracted work no more often than matched controls. A lexical AI score appears to separate the literature roughly hundredfold on the same outcome. That separation is article length: flagged papers carry 44.7 references against 9.2, and the proxy survives no reasonable specification once reference count is controlled. No per-paper design we can build attributes the population shift to AI assistance, and an excess-vocabulary proxy now used to compare LLM prevalence across fields and journals does not survive the obvious control. Data, code, and the artifact corpus are available at `https://github.com/garyzhang1006/ai-science-zombie-data`.

1 Introduction

Retraction is the scientific record's error-correction mechanism, and it works only if authors check the current status of what they cite. Large language models complicate that mechanism in a specific, foreseeable way: a model whose training data was collected before a retraction has no representation of it, and a model asked to summarize a field will reproduce the retracted claim with the same fluency as any other. The chatbot-level evidence is already troubling, since a recent audit found that ChatGPT drew on 61.5% of retracted stem-cell articles when answering domain questions and mentioned the retraction in only 41.7% of those responses (Zhang et al., 2025), and a broader study reached similar conclusions across fields (Thelwall et al., 2025). What no study has measured is whether the published literature itself now cites retracted work at an elevated rate because of AI-assisted writing, which is the question this paper answers.

The measurement problem is harder than it looks, and the difficulty shapes our design. Citations to retracted papers are rare, AI involvement is observed only through noisy proxies, and a naive regression of zombie-citation rates on a lexical AI score is attenuated by proxy error, so a null from that regression would describe the proxy rather than the literature. We therefore built the study around three prongs whose failure modes do not overlap. The first is a population interrupted time

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series, which needs no per-paper attribution at all. The second is a cohort in which AI involvement is near-certain, because the papers carry verbatim LLM artifacts such as "as an AI language model" that usually reach print only through unedited pasting. The third makes the lexical score of Kobak et al. (2024) and Liang et al. (2025) honest instead of discarding it, using partial identification under misclassification (Manski, 2003).

Running all three was worth the cost, because they disagree. The population series moves at the release of ChatGPT, while the cohort with certain exposure does not move at all, and the lexical AI score separates sharply until reference count is controlled and then barely at all. A study built on any one prong would have reported a clean result and been wrong about what it measured.

In the same cohorts we track a second, denser outcome: references that do not resolve, which prior evidence links to model fabrication far above the human baseline (Walters and Wilder, 2023; Bhattacharyya et al., 2023). A zombie citation is real-but-refuted knowledge propagating; an unresolvable reference is fabricated support.

We contribute four things, the first being a population-level measurement of zombie-citation dynamics across the LLM transition, covering 204.9 million citing works and 216,514 zombie citation events. The second is the artifact cohort itself, a public corpus of 1,731 papers with certain LLM involvement and matched controls, with a measured false-positive rate for every phrase we use. The third is a set of bounded rather than attenuated estimates of the dose-response between lexical AI involvement and citation integrity. The fourth is a warning: that proxy's apparent association with citation integrity is almost entirely reference count, and what remains does not hold across specifications.

2 Related work

Post-retraction citation is a longstanding integrity concern predating language models, covering the persistence of citations to retracted work and interventions such as the RISRS recommendations (Schneider et al., 2022). Case studies have traced the citation network of a single prominent retracted paper and found direct citation to be the main propagation channel (van der Vet and Nijveen, 2016). We add scale and a treatment contrast, measuring the phenomenon across the full indexed literature and asking whether the LLM transition changed its rate.

On the AI side, audits of conversational models document the retraction blindness described above (Zhang et al., 2025; Thelwall et al., 2025). They also document fabricated references, with Walters and Wilder measuring 55% of GPT-3.5 citations and 18% of GPT-4 citations as fabricated, alongside substantive errors in a further 24% of the GPT-4 citations that do point at real work (Walters and Wilder, 2023; Bhattacharyya et al., 2023). Strzelecki catalogued verbatim artifact phrases across premier journals (Strzelecki, 2025), establishing that unedited model output reaches print; we industrialize that into a matched-cohort design. Prevalence estimation (Kobak et al., 2024; Liang et al., 2025; Thelwall, 2025) supplies our dose proxy, and to our knowledge no prior work connects these threads to measured citation-integrity outcomes.

3 Data and design

3.1 Retraction and citation data

Retraction events come from the Retraction Watch database, which yields 60,247 retractions carrying both a usable DOI and a retraction date, of which 59,924 (99.5%) match a work in OpenAlex. Citation events and reference lists come from the OpenAlex parquet snapshot of 26 June 2026. We pin that single release so that every number here is reproducible against a frozen corpus rather than a moving index. We work from the snapshot rather than the REST API by necessity: the API now meters usage at 1,000 calls or $0.10 per day on its free tier, which cannot cover a scan of half a billion works, and the snapshot is unmetered.

A citation is a zombie citation if the citing paper appeared at least six months after the cited paper's retraction date, the buffer absorbing indexing lag. We exclude citing works that mention retraction in their title or abstract, since citing a retracted paper to discuss its retraction is the correction system working as intended. Scanning all 2,446 snapshot files yields 216,514 zombie citation events across

Table 1: Artifact phrase registry with false-positive rates measured against the pre-ChatGPT era. A phrase appearing in a paper published before 2023 cannot be unedited model output, so the pre-2023 share of hits estimates the phrase's false-positive rate.

Phrase	Hits	Pre-2023	FP rate	Status
as an AI language model	9,131	111	1.2%	retained
as a large language model, I	244	1	0.4%	retained
as of my last knowledge update	157	3	1.9%	retained
as an AI developed by OpenAI	99	3	3.0%	retained
my knowledge cutoff	89	0	0.0%	retained
I don't have access to real-time	22	1	4.5%	retained
I cannot browse the internet	3	0	0.0%	retained
Regenerate response	9,085	7,926	87.2%	excluded

204.9 million citing works published between 2018 and 2026. Both the numerator and the denominator come from that same pass, which is why the monthly rate needs no sampling correction.

3.2 The artifact cohort

We harvest candidate artifact papers by full-text search over a registry of verbatim LLM phrases. This step needs more care than it first appears, and getting it wrong would have destroyed the cohort. OpenAlex full-text search is stemmed, so the phrase "regenerate response" matches "regenerative response" and returns 9,085 hits dominated by zebrafish and stem-cell regeneration papers. Those hits would have made up 82% of our cohort had we kept the phrase.

We caught this with a negative control that the study period supplies for free. No paper published before 2023 can contain unedited ChatGPT output, so every pre-2023 hit is a false positive by construction, and the pre-2023 share of a phrase's hits estimates its false-positive rate directly. Table 1 reports that rate for every phrase we screened, and "regenerate response" fails at 87.2%, so we exclude it. The seven surviving phrases range from 0.0% to 4.5%, with the workhorse phrase "as an AI language model" at 1.2% across 9,131 hits. This converts the claim that these phrases make model involvement certain from an assertion into a measurement.

Filtering removes papers whose own subject is language models, since those quote the phrases deliberately: 3,613 candidates are dropped on title, 1,868 on subfield, and 1,549 for carrying no references. The filter is imperfect: on a seeded 100-paper sample of the retained cohort its precision is 70% (61 to 79), the misses being AI-topic papers where the phrase may be quotation. What remains is 1,731 papers, all published in 2023 or later, of which 1,643 carry a venue identifier. We match each to five controls from the same venue and year, within 25% of its reference-list length and nearest in citation count, giving 7,644 matched pairs. Reference-list lengths come out nearly identical across arms, 42.55 against 42.45, so the matching binds. Within this cohort the exposure is close to observed rather than proxied, at the price of representing only the least careful tail of AI-assisted writing, a selection we return to in Section 6.

3.3 Outcomes, series, and bounds

For every paper we compute two outcomes over its reference list. The first is the count of zombie citations as defined above; the second counts unresolvable references, meaning entries whose DOI fails to resolve in Crossref and whose bibliographic string matches no indexed work under fuzzy search at a similarity threshold of 85. The population series aggregates the first outcome monthly within field strata, and the interrupted time series allows a level and slope change at the ChatGPT release in December 2022, with Newey-West standard errors and placebo breaks at each month of 2019 through 2021 calibrating the procedure's own false-positive rate.

The bounding analysis follows Manski (2003). Given a lexical AI score with assumed sensitivity and specificity, the observed outcome-by-score table set-identifies the true outcome rates among AI-assisted and unassisted papers. We solve this over a grid rather than at any single assumed point, and the data disciplines the grid. Our rule flags 0.86% of 2023-onward papers, and the prevalence solution requires specificity above $1 - P(S{=}1) = 0.991$. Any specificity below that admits more

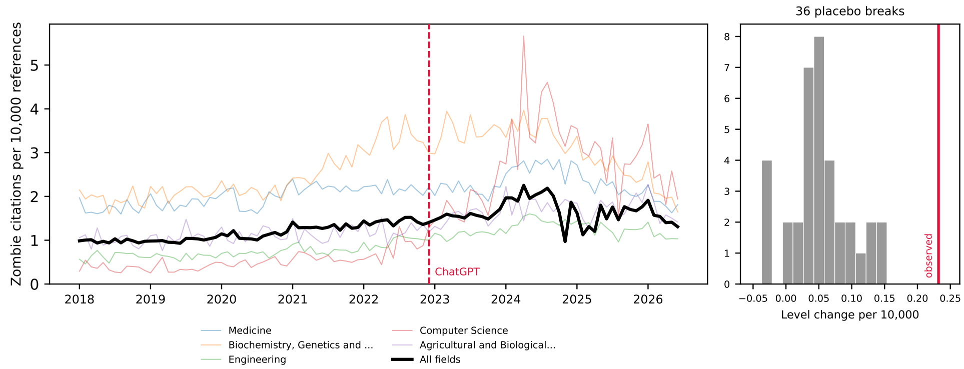

Figure 1: Zombie citations per 10,000 references, monthly, 2018–2026, for the five fields contributing the most events and for the literature as a whole. The vertical line marks the ChatGPT release; the inset shows the distribution of 36 placebo break estimates from 2019–2021 against the estimated 2022 break.

false positives than observed positives, and so no solution at all. A rule demanding two distinct markers from a 37-term list is high-specificity and low-sensitivity by construction, and the grid sits in that regime.

The outcome for this prong is a per-paper indicator, whether a paper carries at least one zombie citation, and that indicator rises with the length of a paper's reference list. The lexical rule needs two markers inside an abstract, so it selects papers with long abstracts, which are full research articles carrying many references. Across the unrestricted sample, flagged papers average 44.7 references against 9.2 for the rest. We therefore compute the bounds within reference-count strata, and take papers with at least 20 references as the primary specification, which compares full articles to full articles at 61.2 references against 50.8.

4 Results

We take the three prongs in order of how much they assume. The population series assumes nothing about individual papers, the artifact cohort observes exposure directly, and the lexical AI score infers it.

4.1 The population rate shifts at the ChatGPT release

The population rate of zombie citation runs at 1.160 per 10,000 references over the pre-period and rises through it, and the interrupted time series estimates a level change of $+0.232$ at the ChatGPT release (standard error 0.130, $p = 0.074$), a 20% increase on the pre-period rate. The conventional p-value is marginal, but it is the weaker of the two tests available here. Across 36 placebo breaks placed through 2019 to 2021, the largest absolute level change is 0.154 and their standard deviation is 0.047, so the real break sits five placebo standard deviations out and exceeds every placebo the procedure produces on data where nothing happened, a placebo-based p below $1/37$ (Figure 1). Placebo fits use only pre-break months, so they run on shorter series than the real fit, which widens their spread and makes the comparison conservative.

The level change is stable under truncation: refitting with the series ending in June 2025, December 2025, March 2026, and June 2026 gives $+0.193$, $+0.190$, $+0.194$, and $+0.232$. The slope change is not. It reads -0.0116 ($p = 0.005$) on the full series but decays to -0.0086 ($p = 0.312$) when the last year is dropped, and the snapshot's final months are visibly under-indexed, carrying 9.0 million references in June 2026 against a monthly norm near 15 million. We therefore report the level shift as the population result and treat the post-break decline as an artifact of the indexing tail.

The break concentrates where one would expect AI writing tools to have landed first. Computer Science shows a level change of $+1.489$ ($p = 0.006$). Biochemistry follows at $+0.606$ ($p = 0.043$), then Engineering at $+0.296$ ($p = 0.010$), Agricultural and Biological Sciences at $+0.297$ ($p =$

Table 2: Artifact cohort against matched controls. Rates per 1,000 references; ratios with 95% cluster-bootstrap intervals resampling papers. The unresolvable-reference row rests on 7,873 papers and 415,750 references checked against Crossref.

Outcome	Artifact papers	Matched controls	Ratio
Zombie citations	0.157	0.179	0.878 [0.379, 1.632]
Unresolvable references	138.84	127.74	1.087 [1.005, 1.174]
Reference-list length	42.55	42.45	

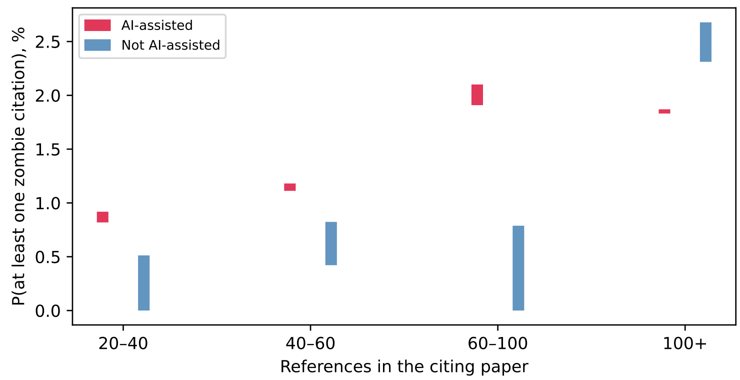

Figure 2: Set-identified probability of carrying at least one zombie citation, by lexical AI score, within strata of reference-list length. Bars span the full range of solutions across the grid. Each flagged bar rests on 6 to 11 events, so the strata illustrate the direction of the adjustment rather than establishing a dose-response.

0.005), and Materials Science at $+0.174$ ($p = 0.017$). Medicine, the largest field by volume, does not move ($+0.163$, $p = 0.362$). Eight fields were tested, and Computer Science ($p = 0.0057$) and Agricultural and Biological Sciences ($p = 0.0045$) are the two that survive a Bonferroni threshold of 0.00625; we read the rest as suggestive of a computational concentration rather than as separately established.

4.2 Near-certain LLM involvement predicts nothing

Within the cohort where LLM involvement is near-certain, the zombie-citation result is a null. Artifact papers cite retracted work at 0.157 per 1,000 references against 0.179 among controls, a ratio of 0.878 whose 95% interval, 0.379 to 1.632, comfortably contains parity (Table 2). This is a bounded null rather than an absence of evidence: the design rules out effects above roughly 1.6 times the control rate, and the point estimate sits below one. (A cohort this size could not have detected a 20% effect, which is the size the population series reports.) The endpoint is sparse. At 0.157 per 1,000 references, 1,645 artifact papers carrying 42.55 references each yield about eleven zombie citations in total.

The denser endpoint carries what statistical weight the cohort has, as the design anticipated. Artifact papers carry 138.84 unresolvable references per 1,000 against 127.74 among controls, a ratio of 1.087 with an interval of 1.005 to 1.174. The effect is real but small, and its interval touches parity closely enough that we decline to describe it as robust. Its absolute level also deserves care: an unresolvable rate near 13% across both arms is far too high to represent fabrication, and mostly reflects Crossref indexing gaps for books, theses, and items without DOIs. Only the ratio between matched arms is interpretable, and only under the assumption that those indexing gaps fall symmetrically across artifact and control papers drawn from the same venue and year.

4.3 The lexical AI score measures article length

Applied without restriction to a seeded sample of 629,900 papers from 2023 onward, this prong gives the study's largest contrast and a spurious one. The rule flags 0.86% of papers. Among those, 1.057% carry at least one zombie citation against 0.113% among the rest, and the correction set-identifies [1.24%, 5.11%] against [0.00%, 0.089%], intervals that never overlap on the grid. The comparison is about article type. Flagged papers average 44.7 references against 9.2, because a rule needing two markers inside an abstract selects long-abstract full articles, and the outcome counts whether a paper carries any zombie citation at all.

Restricting to papers with at least 20 references brings the arms to 61.2 against 50.8, and the naive contrast falls from roughly ninefold to 1.55 (1.261% against 0.816%). Stratifying leaves flagged papers higher in three of four strata and lower in the fourth (Figure 2), but those cells rest on 6 to 11 flagged events, and the reversal above 100 references is 6 events against 105 at a Fisher p of 0.84. We report the strata for the shrinkage across them and decline to read a dose-response into them.

The settling test fits the event indicator on the proxy and log reference count together, across 78,465 papers and 652 events. Reference count dominates at an odds ratio of 2.21 per log unit ($p = 2 \times 10^{-38}$), against 1.35 for the proxy (95% interval 0.95 to 1.92, $p = 0.097$). On the full sample the proxy looks significant only because reference count there spans 1 to several hundred.

Sweeping both thresholds shows how little the proxy contributes. Across twelve specifications, crossing marker thresholds of one to three with reference cuts at 10, 20, 30, and 40, the adjusted odds ratio stays between 1.20 and 1.74 and eight intervals contain one. The four that reach significance are those with the loosest reference cut, where the most short papers remain and residual length confounding is largest. We read this as a small association that no specification test establishes, which is far from the hundredfold separation the unconditioned fit reports.

This is the paper's most transferable result. Excess-vocabulary proxies already compare prevalence across subpopulations: Kobak et al. (2024) report a lower bound that "differed across disciplines, countries, and journals, reaching 40% for some subcorpora." Any such comparison inherits whatever else the markers track, and pairing those estimates with an outcome inherits it twice. Misclassification bounds do not protect against this. The correction assumes the proxy errs at random with respect to the outcome, and length-driven selection violates that in the direction that inflates the result.

5 Discussion

The three prongs give three different answers, and the disagreement is the paper's substance. The population rate moves at the ChatGPT release by an amount that survives every placebo the data can generate, and the move concentrates in the fields where AI writing tools arrived first. The cohort with near-certain exposure shows nothing at all: its zombie endpoint sits below parity, and its denser endpoint clears parity by a margin a reviewer may call fragile. The lexical AI score seems to split the literature, then holds no specification-robust signal once reference count enters the model.

The three now agree on the per-paper question, since neither observed nor inferred exposure predicts citing retracted work, and the design that appeared to disagree was measuring article length rather than writing practice. That is the failure mode the bounds machinery was built to expose and could not, because misclassification correction assumes the proxy errs at random with respect to the outcome. The cohort where exposure is observed is where a causal effect should be largest, and it is absent there. What the proxy separates is a population differing from the rest of the literature in more ways than its use of language models, above all in how much it cites.

This leaves the population result standing and unexplained, which we think is the honest place to leave it: the break concentrates in computational fields, survives 36 placebo breaks and four endpoints, and coincides with the arrival of AI writing tools. It is equally consistent with contemporaneous changes in indexing, in citation practice, or in who published in those fields in 2023. Separating these needs per-paper exposure that is neither self-reported nor lexical, a gap public data cannot close.

The cohort null needs one qualification, since it is selected on carelessness and therefore bounds the careless tail rather than AI assistance at large. That tail is nonetheless where integrity screening would start, which makes the null useful to anyone planning such a programme.

For the workshop's framing question, the study offers a measured instance of a general worry and a sharper caution about how the worry gets measured. The excess-vocabulary proxies now in wide circulation would have answered it confidently and, on this evidence, wrongly.

6 Limitations

The unresolvable-reference comparison queried every one of the 9,009 intended papers, of which 7,873 (87.3%) deposit reference data in Crossref at all; the remainder are absent from the comparison for a reason unrelated to their content. OpenAlex full-text coverage is partial and skewed toward open access, so the artifact cohort undercounts paywalled carelessness. The artifact filter's precision is 70% (61 to 79) on a seeded 100-paper sample, adjudicated by a language model on title and abstract rather than by a human on full text. Using a model to audit a model-artifact filter is not independent, and we report the figure as indicative. The era test bounds how often a phrase fires on text no model could have written. It says nothing about a 2023-or-later author quoting the phrase while discussing model output, which the topic and title filters target instead. We publish the full candidate list with every drop reason, together with a seeded 100-paper sample, so that any reader can audit both. Retraction Watch coverage is itself uneven across fields and eras. More seriously, the interrupted time series identifies a break at the ChatGPT release but cannot by itself attribute it to LLM writing rather than to contemporaneous changes in indexing or citation practice, which the placebo distribution mitigates without eliminating. The bounds correct for misclassification of the lexical AI score and not for confounding, and the conditioned estimate holds reference count fixed but nothing else, so field, venue, and author language remain uncontrolled. Finally, all three prongs measure citation to retracted work, and none measures whether the citing text endorses the retracted claim, so our rates are upper bounds on endorsement.

7 Conclusion

The literature's zombie-citation rate rose about 20% when AI writing tools arrived, against a pre-period rate of 1.160 per 10,000 references, and the rise concentrates in computational fields. The papers we can show a language model wrote display no such elevation, which leaves the population result standing without a mechanism we can name, and we would rather report that gap than close it by picking whichever prong tells the cleaner story. The measurement lesson outlasts this particular question, because an excess-vocabulary proxy of the kind now used to compare LLM prevalence across disciplines separated the literature roughly hundredfold on citation integrity. None of that separation survived a control for how many references a paper carries. The artifact cohort with its measured per-phrase false-positive rates, the bounds machinery, and the monthly series are released at `https://github.com/garyzhang1006/ai-science-zombie-data`.

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A Artifact phrase registry and filtering

Table 1 gives the full registry with raw counts and measured false-positive rates. The filter drops candidates in a fixed order and records a reason for each. Phrase retired for stem collision accounts for 9,088, published before 2023 for 119, and subject classified in an AI subfield for 1,868. A title matching a language-model topic pattern drops 3,613, an abstract carrying two or more meta-discussion terms drops 363, a document type outside the article, review, chapter, preprint, and dissertation set drops 531, and an empty reference list drops 1,549. These counts sum to the 18,862 unique candidates; 1,731 survive. The complete candidate table with per-paper decisions ships in the repository as `artifact_candidates.csv`, and a seeded 100-paper sample for hand validation ships as `hand_validation_sample.csv`.

B Robustness and practical considerations

Section 4 refits the series at four endpoints, where the level change varies between $+0.190$ and $+0.232$ while the slope change loses significance. The six-month buffer, the 25% matching tolerance, and the fuzzy-match threshold of 85 live in `config.py`; `bounds.json` reports the unrestricted fit, the primary fit, and every stratum.

Four decisions in this pipeline each cost a day and changed a number in this paper. We record them because the errors are specific to citation-scale measurement.

Stemmed full-text search collides with domain vocabulary. Our registry originally included "regenerate response," which OpenAlex stems and matches against "regenerative response." It returned 9,085 hits, 87.2% of them predating ChatGPT, drawn from zebrafish and stem-cell regeneration work, and it was 82% of the cohort before removal. The era test in Table 1 caught it, and every phrase now passes that test before admission.

OpenAlex serves abstracts as an inverted index, not as text. Abstracts arrive as JSON mapping each word to its positions, so scoring the serialized string matched no markers: they appear as quoted dictionary keys. The failure is silent, returning a proxy-positive rate of zero and unidentified bounds rather than an error. Parsing the index first gives 0.86%.

A per-paper outcome indicator inherits reference-list length. Our first bounds specification asked whether a paper carried at least one zombie citation, which rises mechanically with the number of references. Because the lexical rule needs two markers inside an abstract, it selects long-abstract full articles carrying 44.7 references against 9.2 for the rest. The unconditioned contrast was roughly a hundredfold because the arms differed in length, and the conditioned one is not significant. We now put length on the right-hand side of any per-paper indicator over reference lists.

Debug aggregates merge silently with production shards. Partial results carry an `<i>of<n>` tag, and a test run left a stray `2130of2446` beside the six production shards. Stage 4 globbed both, since the pattern matched either, double-counting that file and shifting the pre-period rate by 0.003. The merge now takes only the largest complete sharding.